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CBSE CLASS 10 | MATHEMATICS

REAL NUMBERS

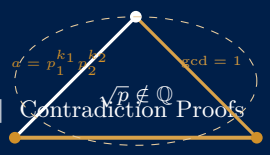
Fundamental Theorem of Arithmetic, Divisibility & Irrationality Proofs

Academic Year 2026–27

Master the Primes. Unpack Unique Factorization.
Prove Irrationality with Absolute Logical Rigor.

Fundamental Theorem of Arithmetic | Exponential HCF/LCM | Divisibility Lemma | Contradiction Proofs

A publication-quality Maths Manthan learning and revision resource



What this handbook includes

- Set-theoretic foundations of $\mathbb{R}$: $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ and the Real Line continuum.
- The Fundamental Theorem of Arithmetic (FTA) and Canonical Prime Decomposition ($n = \prod p_i^{a_i}$).
- The Exponential Power Law of HCF (min) and LCM (max).
- The Dual-Number Product Theorem HCF $\times$ LCM $= a \times b$ and why it strictly fails for three numbers.
- Real-world word problems decision framework (HCF vs LCM).
- Complete formal proofs for irrationality of $\sqrt{2}, \sqrt{3}, \sqrt{5}$ and linear combinations ($a \pm b\sqrt{c}$).

How to use it for maximum marks

In irrationality proofs, always explicitly state that a and b are **coprime integers with** $b \neq 0$ ($\gcd(a, b) = 1$) to avoid a mandatory 1-mark deduction. In word problems, state the prime factorizations in exponential index notation before applying min or max powers. Never write $\text{HCF}(a, b, c) \times \text{LCM}(a, b, c) = a \cdot b \cdot c$ for 3 numbers.

Official syllabus scope (CBSE Class X 2026–27)

The published CBSE Class X curriculum for 2026–27 prescribes **Fundamental Theorem of Arithmetic – statements after reviewing work done earlier and after illustrating and motivating through examples, Proofs of irrationality of $\sqrt{2}, \sqrt{3}, \sqrt{5}$.**
Strictly Excluded Topics (Deleted from Rationalized Syllabus):
Section 1.2: Euclid's Division Lemma ($a = bq + r$) and its algorithmic application to find HCF are **completely deleted**. Questions representing integers as $6q+1$, $6q+3$ or proving squares of forms $(3m)^2$ are excluded. Section 1.4: Revisiting Rational Numbers and Their Decimal Expansions (terminating criteria $q = 2^n 5^m$) is **completely deleted**.

Source: CBSE Mathematics Class X Curriculum (2026–27), Unit I: Number Systems, Chapter 1.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 1

CORE SYLLABUS

GEOMETRIC & SET FOUNDATIONS

The Number Hierarchy, Real Line Continuum & Decimal Duality

Module 1 | The Set-Theoretic Hierarchy of Numbers

The system of Real Numbers ($\mathbb{R}$) is built hierarchically through successive algebraic extensions:

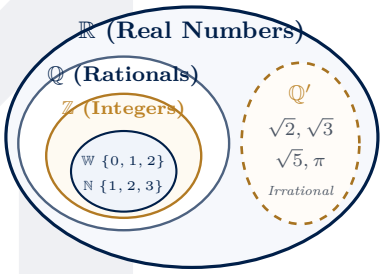
Set Name	Symbol	Formal Mathematical Definition	Structural Role & Intuition
Natural Numbers	$\mathbb{N}$	$\{1, 2, 3, 4, 5, \dots\}$	Primitive discrete counting elements.
Whole Numbers	$\mathbb{W}$	$\{0\} \cup \mathbb{N} = \{0, 1, 2, 3, \dots\}$	Introduces the additive identity (null quantity).
Integers	$\mathbb{Z}$	$\{\dots, -2, -1, 0, 1, 2, \dots\}$	Introduces additive inverses (negative quantities).
Rational Numbers	$\mathbb{Q}$	$\left\{ \frac{p}{q} \,\middle\vert\, p, q \in \mathbb{Z},\ q \neq 0,\ \gcd(p, q) = 1 \right\}$	Closed under division; expresses all fractional ratios.
Irrational Numbers	$\mathbb{Q}' \ / \ \mathbb{I}$	$\{x \in \mathbb{R} \mid x \notin \mathbb{Q}\}$	Numbers that cannot be written as a fraction of integers.
Real Numbers	$\mathbb{R}$	$\mathbb{Q} \cup \mathbb{Q}'$	Complete, gapless continuous geometric number line.

Module 2 | Decimal Representation Duality

Decimal Characterization of Real Numbers

Every real number possesses a unique decimal expansion:

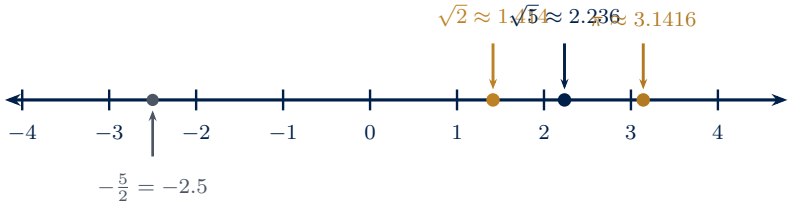
- **Rational ($\mathbb{Q}$):** Decimal expansion is either:
  (a) **Terminating:** e.g., $\frac{3}{8} = 0.375$.
  (b) **Non-terminating Recurring (Periodic):** e.g., $\frac{1}{3} = 0.333\cdots = 0.\overline{3}$, $\frac{1}{7} = 0.\overline{142857}$.
- **Irrational ($\mathbb{Q}'$):** Decimal expansion is **strictly non-terminating and non-recurring**.
  - $\sqrt{2} = 1.41421356237\dots$ (No repeating block).
  - $\pi = 3.14159265358\dots$ (Cannot be matched by any ratio).



Module 3 | The Real Number Line Continuum (Dedekind–Cantor Completeness)

The Fundamental Geometric Continuum Axiom

The Dedekind–Cantor Axiom: To every real number there corresponds a **unique point** on the number line, and conversely, to every point on the number line there corresponds **exactly one real number**. The rationals $\mathbb{Q}$ are dense but full of holes; the irrationals $\mathbb{Q}'$ fill every single geometric gap.



CHAPTER 1

CORE SYLLABUS

FUNDAMENTAL THEOREM OF ARITHMETIC

Prime Building Blocks, Uniqueness & Canonical Representation

Module 1 | The Fundamental Theorem of Arithmetic (FTA)

Formal Statement of the Theorem (Carl Friedrich Gauss, 1801)

Theorem: *Every composite number can be expressed (factorized) as a product of primes, and this factorization is* ***unique****, apart from the order in which the prime factors occur.*

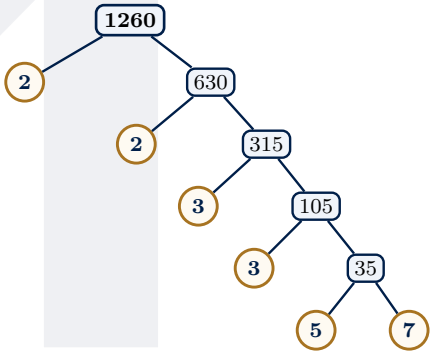
- **Existence Clause:** Any integer $n > 1$ can be broken down into a product of prime building blocks.
- **Uniqueness Clause:** There is only **one singular set of prime powers** that produces n. No matter which method or branching sequence is used, the final prime components are identical!

Module 2 | Canonical Prime Decomposition

Writing primes in ascending order gives the **Canonical Representation** of an integer n:

$$\mathbf{n = p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_k^{a_k} = \prod_{i=1}^{k} p_i^{a_i}}$$

where $p_1 < p_2 < p_3 < \cdots < p_k$ are distinct positive primes and $a_i \in \mathbb{N}$ are their respective natural multiplicities.



$1260 = 2^2 \times 3^2 \times 5 \times 7$

Module 3 | The Factor Tree Algorithm

A **Factor Tree** is a visual recursive algorithm for obtaining canonical prime decomposition:

1. Write the composite number n at the root.
2. Branch n into its smallest prime divisor p and quotient q.
3. Circle the prime factor p (leaf node: terminated branch).
4. If quotient q is composite, repeat the branching process.
5. Termination occurs when all leaf nodes are prime numbers.
6. Collect identical prime leaves into exponential powers.

Worked Model: Finding Unknown Missing Branches in a Factor Tree

Board Problem: In a given factor tree, a parent node x splits into 3 and y, and y splits into 5 and 13. Determine the values of x and y.
Solution by Inversion:
Step 1: Compute child node y: $y = 5 \times 13 = \mathbf{65}$.
Step 2: Compute parent node x: $x = 3 \times y = 3 \times 65 = \mathbf{195}$.
Verification: Canonical prime factorization of $195 = 3 \times 5 \times 13$. Uniqueness guarantees singular solution.

CHAPTER 1

CORE SYLLABUS

THE EXPONENTIAL LAW OF HCF & LCM

Min/Max Power Protocols & The Two-Number Product Theorem

Module 1 | Exponential Definitions of HCF and LCM

Let the canonical prime factorizations of two natural numbers a and b be expressed across all involved primes:

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k} \qquad \text{and} \qquad b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k} \quad (\alpha_i, \beta_i \geq 0)$$

The Fundamental Power Laws for HCF and LCM

- **Highest Common Factor (HCF):** Product of the **smallest powers** of each common prime factor:

$$\text{HCF}(\mathbf{a}, \mathbf{b}) = \prod_{\mathbf{i=1}}^{\mathbf{k}} \mathbf{p_i}^{\min(\alpha_i, \beta_i)} = \mathbf{p_1}^{\min(\alpha_1, \beta_1)} \cdot \mathbf{p_2}^{\min(\alpha_2, \beta_2)} \cdots \mathbf{p_k}^{\min(\alpha_k, \beta_k)}$$

- **Lowest Common Multiple (LCM):** Product of the **greatest powers** of each prime factor involved:

$$\text{LCM}(\mathbf{a}, \mathbf{b}) = \prod_{\mathbf{i=1}}^{\mathbf{k}} \mathbf{p_i}^{\max(\alpha_i, \beta_i)} = \mathbf{p_1}^{\max(\alpha_1, \beta_1)} \cdot \mathbf{p_2}^{\max(\alpha_2, \beta_2)} \cdots \mathbf{p_k}^{\max(\alpha_k, \beta_k)}$$

Module 2 | The Dual-Number Product Theorem

Theorem: Product Law for Two Numbers

For any two positive integers a and b:

$$\text{HCF}(\mathbf{a}, \mathbf{b}) \times \text{LCM}(\mathbf{a}, \mathbf{b}) = \mathbf{a} \times \mathbf{b}$$

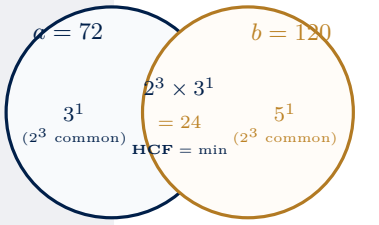
Mathematical Proof via Exponents:
For every prime factor p_i, the sum of exponents is:

$$\min(\alpha_i, \beta_i) + \max(\alpha_i, \beta_i) = \alpha_i + \beta_i$$

Taking the product across all primes:

$$\prod p_i^{\min(\alpha_i, \beta_i)} \times \prod p_i^{\max(\alpha_i, \beta_i)} = \prod p_i^{\alpha_i + \beta_i} = a \times b$$

$$\therefore \text{LCM}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \times \mathbf{b}}{\text{HCF}(\mathbf{a}, \mathbf{b})}, \qquad \text{HCF}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \times \mathbf{b}}{\text{LCM}(\mathbf{a}, \mathbf{b})}$$



Total Union = LCM $= 2^3 \times 3^2 \times 5 = 360$

Module 3 | The Three-Number Fallacy (CBSE Lethal Trap)

Critical Warning: Product Law Fails for Three Numbers!

CBSE Board Trap: It is **STRICTLY FALSE** that $\text{HCF}(a, b, c) \times \text{LCM}(a, b, c) = a \times b \times c$!
Concrete Counterexample: Let $a = 6 = 2 \times 3$, $b = 8 = 2^3$, $c = 12 = 2^2 \times 3$.

- $\text{HCF}(6, 8, 12) = 2^1 = \mathbf{2}$; $\quad \text{LCM}(6, 8, 12) = 2^3 \times 3 = \mathbf{24}$.
- LHS $= \text{HCF} \times \text{LCM} = 2 \times 24 = \mathbf{48}$.
- RHS $= a \times b \times c = 6 \times 8 \times 12 = \mathbf{576}$. $\quad$ **Clearly,** $48 \neq 576$**!**

True Formula for 3 Numbers (For Reference):

$$\text{LCM}(a, b, c) = \frac{a \cdot b \cdot c \cdot \text{HCF}(a, b, c)}{\text{HCF}(a, b) \cdot \text{HCF}(b, c) \cdot \text{HCF}(c, a)}$$

CHAPTER 1

WORKED MODEL

WORD PROBLEMS DECISION ENGINE

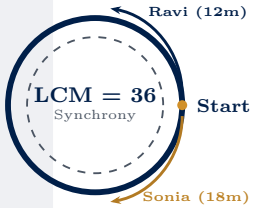
Semantic Triggers, Recurrence Cycles & Optimal Grouping

Module 1 | The Semantic Trigger Decision Framework

When confronted with real-world situational word problems, identify the governing mathematical structure:

Diagnostic Criteria	HCF Model (Subdivision / Partition)	LCM Model (Multiple / Recurrence)
Physical Objective	Dividing larger quantities into smaller, equal groups with **no remainder**.	Finding a common cycle, synchronization point, or smallest expanded quantity.
Semantic Keywords	**Maximum**, largest, greatest, highest capacity, longest tape, equal heaps, minimum number of stacks.	**Minimum**, least, smallest, bells chiming together, traffic lights changing, runners circling a track.
Size Constraint	The result is **smaller than or equal to** given values: HCF $\leq \min(a, b)$.	The result is **larger than or equal to** given values: LCM $\geq \max(a, b)$.

Module 2 | Worked Board Archetype 1: Circular Sports Track (LCM Model)



Problem: Sonia takes 18 min and Ravi takes 12 min for one round of a sports field. Starting together in the same direction, after how many minutes will they meet again at the starting point?
Solution:
Step 1: Model. They meet at the starting point at the **Least Common Multiple** of their lap times: LCM$(18, 12)$.
Step 2: Prime factorizations. $18 = 2 \times 3^2, \quad 12 = 2^2 \times 3$.
Step 3: Compute LCM.
LCM$(18, 12) = 2^{\max(1,2)} \times 3^{\max(2,1)} = 2^2 \times 3^2 = 4 \times 9 = \mathbf{36}$ minutes.
Conclusion: They will meet again after **36 minutes** (Sonia completes 2 laps, Ravi completes 3 laps).

Module 3 | Worked Board Archetype 2: Maximum Sweet Stacking (HCF Model)

Worked Model: Optimal Subdivision of Burfis

Problem: A sweet seller has 420 kaju burfis and 130 badam burfis. She wants to stack them such that each stack has the same number and they take up the least tray area. Find the maximum number of burfis per stack.
Solution:
Step 1: Model. Least tray area requires largest equal stacks leaving no remainder $\implies$ **Highest Common Factor** HCF$(420, 130)$.
Step 2: Prime factorizations. $420 = 2^2 \times 3 \times 5 \times 7, \quad 130 = 2 \times 5 \times 13$.
Step 3: Compute HCF. HCF$(420, 130) = 2^{\min(2,1)} \times 5^{\min(1,1)} = 2^1 \times 5^1 = \mathbf{10}$ burfis per stack.
Conclusion: Each stack contains **10 burfis**. (Total stacks formed: $\frac{420}{10} + \frac{130}{10} = 42 + 13 = \mathbf{55}$ stacks).

CHAPTER 1

CORE SYLLABUS

TERMINAL ZEROES & COMPOSITES

The 2×5 Divisibility Lock & Factoring Verifications

Module 1 | Terminal Zeroes Analysis (n-th Powers)

A perennial CBSE board exam question tests whether powers of an integer can terminate in the digit 0:

The Fundamental Divisibility Lock for Terminal Zero

Theorem: A natural number ends with the digit 0 (i.e. has a terminal zero) if and only if it is a multiple of 10. Since the prime factorization of 10 is 2×5, any number ending in 0 **MUST contain BOTH 2 and 5** as prime factors in its prime decomposition.

Worked Board Archetype: Check if 6^n Ends in 0

Terminal Zero Key

$10 = 2^1 \times 5^1$

☑ Prime 2 present

☒ Prime 5 missing!

$6^n = 2^n \times 3^n \implies$ No 0

Question: Check whether 6^n can end with the digit 0 for any natural number $n \in \mathbb{N}$.
Rigorous Board Proof:
Step 1: Factor the base.

$$6^n = (2 \times 3)^n = 2^n \times 3^n$$

Step 2: Apply the Uniqueness Clause of FTA. The prime factors of 6^n are only 2 and 3. By the Fundamental Theorem of Arithmetic, this prime factorization is **unique**. There are no other primes in the factorization of 6^n.
Step 3: State the contradiction and conclusion. For 6^n to end with digit 0, it must have 5 as a prime factor. Since 5 is absent from the prime factorization of 6^n, 6^n **cannot end with the digit** 0 **for any** $n \in \mathbb{N}$. ■

Module 2 | Algebraic Verification of Composite Numbers

Definition and Verification Protocol for Composite Numbers

Definition: A number is composite if it has **more than two factors** (i.e. factors other than 1 and itself).
Board Strategy: NEVER expand the full arithmetic product! Instead, factor out the common term to prove that the expression is a product of two or more integers greater than 1.

Archetype 1: Expression with Addend 13

Problem: Explain why $7 \times 11 \times 13 + 13$ is a composite number.
Solution:
Take common factor 13 outside:

$$\begin{aligned} 7 \times 11 \times 13 + 13 &= 13 \times (7 \times 11 + 1) \\ &= 13 \times (77 + 1) \\ &= 13 \times 78 \\ &= 13 \times (2 \times 3 \times 13) \\ &= 2 \times 3 \times 13^2 \end{aligned}$$

Since the given expression can be expressed as a product of prime factors $2, 3, 13$, it has factors other than 1 and itself. Hence, it is a **composite number**. ■

Archetype 2: Factorial Expression + 5

Problem: Explain why $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is composite.
Solution:
Take common factor 5 outside:

$$\begin{aligned} E &= 5 \times (7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1) \\ &= 5 \times (1008 + 1) \\ &= 5 \times 1009 \end{aligned}$$

Since 5 and 1009 are both integers greater than 1, the number E has at least $1, 5, 1009$, and E as distinct positive factors.
By definition of composite numbers, having more than two factors proves that E is a **composite number**. ■

CHAPTER 1

CORE SYLLABUS

THE DIVISIBILITY LEMMA

Theorem: If Prime p Divides a^2, Then p Divides a

Module 1 | Theorem Statement & Critical Mathematical Role

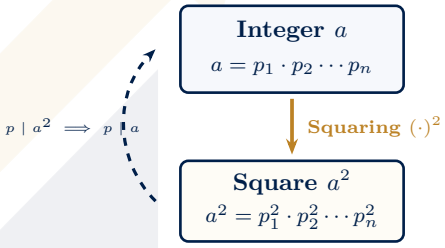
The Pillar of All Irrationality Proofs (Theorem 1.2)

Theorem: *Let p be a prime number. If p divides a^2 (where a is a positive integer), then p divides a.*

$$\mathbf{p \mid a^2 \implies p \mid a} \qquad (\mathbf{p} \text{ is prime}, \mathbf{a} \in \mathbb{Z}^+)$$

Crucial Pedagogical Role: This single lemma is the deductive engine of every square root irrationality proof in the CBSE curriculum. Without it, the contradiction cannot be established!

Module 2 | Complete Formal Proof via FTA



Proof:

Step 1: Declare prime decomposition of a.
Let the prime factorization of a be:

$$a = p_1 p_2 p_3 \cdots p_n$$

where $p_1, p_2, \ldots, p_n$ are primes (not necessarily distinct).

Step 2: Square both sides.

$$a^2 = (p_1 p_2 \cdots p_n)(p_1 p_2 \cdots p_n) = p_1^2 p_2^2 \cdots p_n^2$$

Step 3: Apply the divisibility condition.
We are given that prime p divides a^2 ($p \mid a^2$).

Step 4: Invoke the Uniqueness of FTA.
By the Fundamental Theorem of Arithmetic, the prime factorization of a^2 is **unique**. The only prime factors of a^2 are $p_1, p_2, \ldots, p_n$. Therefore, p must be one of the prime factors $p_1, p_2, \ldots, p_n$.

Step 5: Conclude divisibility of a.
Since $a = p_1 p_2 \cdots p_n$ and $p \in \{p_1, p_2, \ldots, p_n\}$, it follows directly that p **divides** a. ■

Module 3 | Why Primality of p is Strictly Necessary (Counterexamples)

Fatal Error: The Lemma FAILS if p is a Composite Number!

Crucial Distinction: The theorem holds **ONLY** when p is prime. If p is composite, the implication $p \mid a^2 \implies p \mid a$ is **FALSE**!

Counterexample 1: Let $p = 4$ (composite) and $a = 6$.

- $a^2 = 6^2 = 36$. Here 4 divides 36 ($36 = 4 \times 9$).
- But 4 does **NOT** divide 6 ($6/4 = 1.5 \notin \mathbb{Z}$)!

Why does this happen? A composite number can split its prime factors across the two copies of a: $4 = 2 \times 2$, where one 2 divides the first a and the other 2 divides the second a. A prime number is indivisible and cannot split; it must belong entirely to a itself!

CHAPTER 1

WORKED MODEL

PROOFS OF IRRATIONALITY: PURE SURDS

The Contradiction Method & Rigorous Board Template for $\sqrt{5}$

Module 1 | The Reductio ad Absurdum (Proof by Contradiction) Architecture

Four-Step Universal Contradiction Template

1. **Negation Assumption:** Assume the contrary: that $\sqrt{p}$ is rational ($\sqrt{p} = \frac{a}{b}$, with $a, b \in \mathbb{Z}^+, \gcd(a,b) = 1$).
2. **First Divisibility Deduction:** Square both sides to establish $pb^2 = a^2 \implies p \mid a^2 \implies \mathbf{p \mid a}$.
3. **Second Divisibility Deduction:** Substitute $a = pc$ to establish $b^2 = pc^2 \implies p \mid b^2 \implies \mathbf{p \mid b}$.
4. **Contradiction & Verdict:** a and b share common factor p, contradicting $\gcd(a,b) = 1$. Hence, $\sqrt{p}$ is irrational.

Module 2 | Standard Board Exam Model Proof: $\sqrt{5}$ is Irrational

Complete Full-Mark CBSE Solution Protocol

Theorem: Prove that $\sqrt{5}$ is an irrational number.
Proof:
Step 1: State the contradiction hypothesis.
Let us assume, to the contrary, that $\sqrt{5}$ is a rational number.
Therefore, there exist positive integers a and b ($b \neq 0$) such that:

$$\sqrt{\mathbf{5}} = \frac{\mathbf{a}}{\mathbf{b}}, \quad \text{where } a \text{ and } b \text{ are } \textbf{coprime} \text{ (i.e. } \gcd(a,b) = 1).$$

Step 2: Square both sides and deduce divisibility of a.
Cross-multiplying: $b\sqrt{5} = a$.
Squaring both sides:

$$\mathbf{5b^2 = a^2} \qquad \text{— (Equation 1)}$$

Since 5 divides $5b^2$, it follows that 5 **divides** a^2.
By Theorem (since 5 is a prime number), 5 **divides** a. — (Statement A)
Step 3: Express a and deduce divisibility of b.
Since 5 divides a, we can write $\mathbf{a = 5c}$ for some integer c.
Substituting $a = 5c$ into Equation (1):

$$5b^2 = (5c)^2 = 25c^2 \implies \mathbf{b^2 = 5c^2} \qquad \text{— (Equation 2)}$$

Since 5 divides $5c^2$, it follows that 5 **divides** b^2.
By Theorem (since 5 is prime), 5 **divides** b. — (Statement B)
Step 4: Establish the mathematical contradiction.
From Statement A and Statement B, 5 **is a common factor of both a and b.**
This contradicts our initial foundational assumption that a and b are coprime (having no common factor other than 1).
Step 5: Write the mandatory closing conclusion.
This contradiction has arisen because of our incorrect assumption that $\sqrt{5}$ is rational.
Hence, we conclude that $\sqrt{5}$ **is an irrational number.** ■

Module 3 | Universal Transformation for $\sqrt{2}$ and $\sqrt{3}$

To prove $\sqrt{2}$ or $\sqrt{3}$ is irrational, use the **identical template** with the exact numerical replacements:

$$\sqrt{2}: \quad 2b^2 = a^2 \implies 2 \mid a \implies a = 2c \implies b^2 = 2c^2 \implies 2 \mid b \implies \text{Contradiction with } \gcd(a,b) = 1$$

$$\sqrt{3}: \quad 3b^2 = a^2 \implies 3 \mid a \implies a = 3c \implies b^2 = 3c^2 \implies 3 \mid b \implies \text{Contradiction with } \gcd(a,b) = 1$$

CHAPTER 1

WORKED MODEL

PROOFS OF IRRATIONALITY: COMBINATIONS

Linear Sums, Differences, Products & Quotients

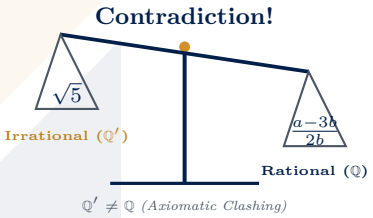
Module 1 | The "Given $\sqrt{p}$ is Irrational" Board Protocol

Algebraic Rearrangement vs. Full Contradiction

CBSE Marking Scheme Rule: When an exam question states: *"Given that $\sqrt{5}$ is irrational, prove that $3+2\sqrt{5}$ is irrational"*:

- Do **NOT** reprove $\sqrt{5}$ from scratch!
- Simply isolate the irrational surd $\sqrt{5}$ on the LHS using basic algebra.
- Show that the RHS consists entirely of rational operations on integers, creating an immediate contradiction between $\mathbb{Q}'$ and $\mathbb{Q}$.

Module 2 | Worked Archetype: $3+2\sqrt{5}$



Problem: Given that $\sqrt{5}$ is irrational, prove that $3+2\sqrt{5}$ is irrational.
Solution:
Step 1: Assumption.
Assume, to the contrary, that $3+2\sqrt{5}$ is rational.
Let $3+2\sqrt{5}=\frac{a}{b}$, where $a,b\in\mathbb{Z}, b\neq 0, \gcd(a,b)=1$.
Step 2: Isolate the surd $\sqrt{5}$.

$$2\sqrt{5}=\frac{a}{b}-3=\frac{a-3b}{b} \implies \sqrt{\mathbf{5}}=\frac{\mathbf{a-3b}}{\mathbf{2b}}$$

Step 3: Analyze rationality.
Since a and b are integers, $a-3b$ and $2b$ are integers with $2b\neq 0$.
Therefore, $\frac{a-3b}{2b}$ is a rational number.
This implies that $\sqrt{5}$ is rational.
Step 4: State contradiction and verdict.
But this contradicts the fact that $\sqrt{5}$ is irrational.
Hence, our assumption is false. $3+2\sqrt{5}$ **is irrational**. ■

Module 3 | Worked Archetypes: Difference & Quotient Forms

Archetype: Difference Form $(5-\sqrt{3})$

Problem: Prove that $5-\sqrt{3}$ is irrational.
Solution:
Let $5-\sqrt{3}=\frac{a}{b}$ $(a,b\in\mathbb{Z}, b\neq 0)$.
Rearrange to isolate $\sqrt{3}$:

$$\sqrt{3}=5-\frac{a}{b}=\frac{5b-a}{b}$$

Since a,b are integers, $\frac{5b-a}{b}$ is rational.
This implies $\sqrt{3}$ is rational, contradicting the known fact that $\sqrt{3}$ is irrational.
$\therefore$ $\mathbf{5-\sqrt{3}}$ is an **irrational number**. ■

Archetype: Quotient Form $(1/\sqrt{2})$

Problem: Prove that $\frac{1}{\sqrt{2}}$ is irrational.
Solution:
Let $\frac{1}{\sqrt{2}}=\frac{a}{b}$ $(a,b\in\mathbb{Z}, a\neq 0, b\neq 0)$.
Inverting both sides:

$$\sqrt{2}=\frac{b}{a}$$

Since $a,b\in\mathbb{Z}$ and $a\neq 0$, $\frac{b}{a}$ is rational.
This implies $\sqrt{2}$ is rational, which contradicts the fact that $\sqrt{2}$ is irrational.
$\therefore$ $\frac{\mathbf{1}}{\sqrt{\mathbf{2}}}$ is an **irrational number**. ■

CHAPTER 1

REFERENCE MATRIX

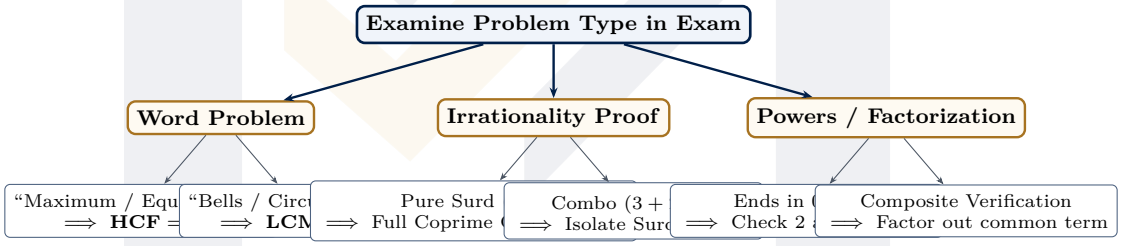
QUICK-REVISION MATRIX

Comprehensive Formula Table, Decision Rules & Board Conventions

Module 1 | Master Formula & Conceptual Rules Matrix

Concept / Theorem	Mathematical Formula	Key Domain Restriction	CBSE Exam Utility
FTA Decomposition	$n = \prod_{i=1}^{k} p_i^{a_i}$	$p_1 < p_2 < \cdots < p_k$ primes, $a_i \geq 1$	Canonical representation.
HCF by Primes	$\text{HCF}(a,b) = \prod p_i^{\min(\alpha_i,\beta_i)}$	Common prime factors only	Subdividing, equal stacks.
LCM by Primes	$\text{LCM}(a,b) = \prod p_i^{\max(\alpha_i,\beta_i)}$	All prime factors involved	Cycles, bells, laps.
Two-Number Law	$\text{HCF}(a,b) \times \text{LCM}(a,b) = a \cdot b$	**Valid ONLY for 2 numbers**	Missing quantity calculation.
Three-Number Rule	$\text{HCF} \times \text{LCM} \neq a \cdot b \cdot c$	Product law strictly fails	Always use prime powers.
Terminal Zeroes	$N = 2^n \times 5^m \times k$ $(n, m \geq 1)$	Requires BOTH 2 and 5 primes	$4^n, 6^n, 12^n$ cannot end in 0.
Composite Proof	$N = k \times m$ $(k, m > 1)$	Has > 2 distinct positive factors	Factor out common addend.
Divisibility Lemma	$p \mid a^2 \implies p \mid a$	p **MUST be prime**	Foundation of surd proofs.
Pure Surd Proof	$\sqrt{p} = a/b \implies pb^2 = a^2$	a, b coprime integers (gcd $= 1$)	Contradiction template.
Linear Combo Proof	$\sqrt{p} = \frac{a-rb}{sb}$	Isolate surd on LHS	LHS $\in \mathbb{Q}'$, RHS $\in \mathbb{Q}$ clash.

Module 2 | Board Question Diagnostic Flowchart



Module 3 | Curriculum Scope Verification (CBSE 2026–27)

Syllabus Status	Curriculum Topic	Action Plan for CBSE Exam
Active Scope	Fundamental Theorem of Arithmetic (FTA) & Canonical Decomposition	Master prime power factorization and factor trees.
Active Scope	Proofs of Irrationality of $\sqrt{2}, \sqrt{3}, \sqrt{5}$ and linear combinations	Write all 5 steps with explicit "coprime" statement.
DELETED	Euclid's Division Lemma ($a = bq + r$) & Long Division HCF	**Do NOT study** (Zero marks allotted).
DELETED	Decimal Expansions of Rationals ($q = 2^n 5^m$ terminating checks)	**Do NOT study** (Completely deleted from syllabus).

CHAPTER 1

EXAM STRATEGY

BOARD STRATEGY & AUDIT

The 5 Fatal Examiner Traps, Arithmetic Hacks & Audit Protocol

Module 1 | The Five Lethal CBSE Board Exam Traps

Examiner Trap Alerts (Where Students Lose Marks)

1. **Trap 1: Omitting "Coprime" in Proof by Contradiction (-1 Mark).** Writing $\sqrt{5} = a/b$ without explicitly stating *"a, b are coprime integers ($b \neq 0$)"* destroys the contradiction premise.
2. **Trap 2: Applying Two-Number Product Law to Three Numbers.** Writing $\text{HCF}(a,b,c) \times \text{LCM}(a,b,c) = a \cdot b \cdot c$ is false! For 3 numbers, always use prime factorization directly.
3. **Trap 3: Multiplying Out in Composite Proofs.** In *"Explain why $7 \times 11 \times 13 + 13$ is composite"*, do NOT multiply to 1014. **Always factor out common addend:** $13(7 \times 11 + 1) = 13 \times 78$.
4. **Trap 4: Confusing HCF and LCM in Context.** If the answer must be smaller than the given numbers (e.g. stack size, container capacity), it is **HCF**. If larger (e.g. time until next bell, laps), it is **LCM**.
5. **Trap 5: Incomplete Closing Contradiction Statement.** Ending abruptly after $5 \mid b$ without stating *"This contradicts coprimality of a, b, which arose from our incorrect assumption..."* loses marks.

Module 2 | Rapid Arithmetic Verification Hacks

MM
100% AUDIT
CBSE 2026–27
Zero Marks Lost

Sanity Checks for Real Numbers

- **Divisibility Invariance:** $\text{LCM}(a,b)$ must be **strictly divisible** by $\text{HCF}(a,b)$. If LCM (mod HCF) $\neq 0$, your calculation has an arithmetic error!
- **Size Bounds:** $\text{HCF}(a,b) \leq \min(a,b) \leq \max(a,b) \leq \text{LCM}(a,b)$.
- **Coprime Rule:** If $\text{HCF}(a,b) = 1$, then $\text{LCM}(a,b) = a \times b$.

Module 3 | The Maths Manthan 5-Point Pre-Submission Audit Protocol

Pre-Submission Verification Checklist

Check	Audit Stage	Verification Action Before Handing in Your Answer Sheet
☐	**1. Coprime Clause**	In every irrationality proof, did I write $\gcd(a,b) = 1$ with $a, b \in \mathbb{Z}^+$ and $b \neq 0$?
☐	**2. Theorem Attribution**	Did I explicitly quote *"By the Fundamental Theorem of Arithmetic"* or *"Since p is prime"*?
☐	**3. Product Theorem Audit**	If calculating for two numbers, did I verify that HCF $\times$ LCM equals $a \times b$?
☐	**4. Terminal Zero Logic**	Did I explicitly show that the prime decomposition lacks the factor 5 (or 2)?
☐	**5. Unit & Wording Check**	In word problems, did I include physical units (minutes, sweets, meters) in the final sentence?

End of Chapter 1 Master Handbook | **Maths Manthan Tuition Classes**

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